Higher-Order Fourier Neural Operator: Additional Experiments and Analysis

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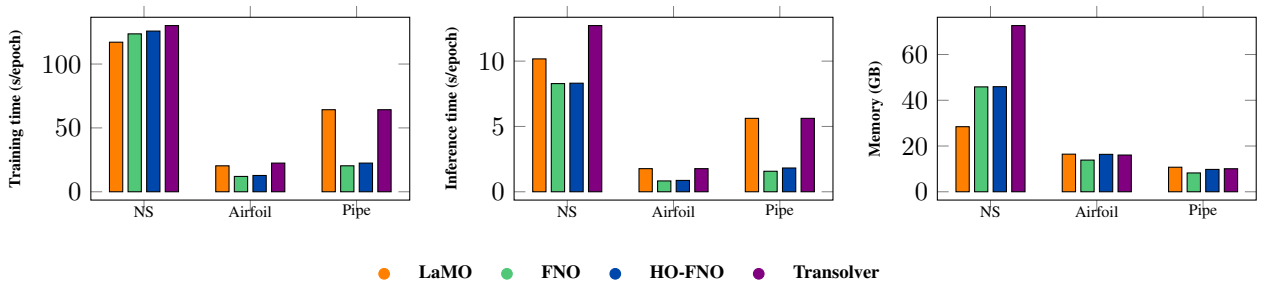

Figure 1. Efficiency comparison on NS, Airfoil, and Pipe on a single Nvidia A100 GPU: (**Left**) training time, (**Middle**) inference time, and (**Right**) memory consumption per epoch. FNO is with improved backbone

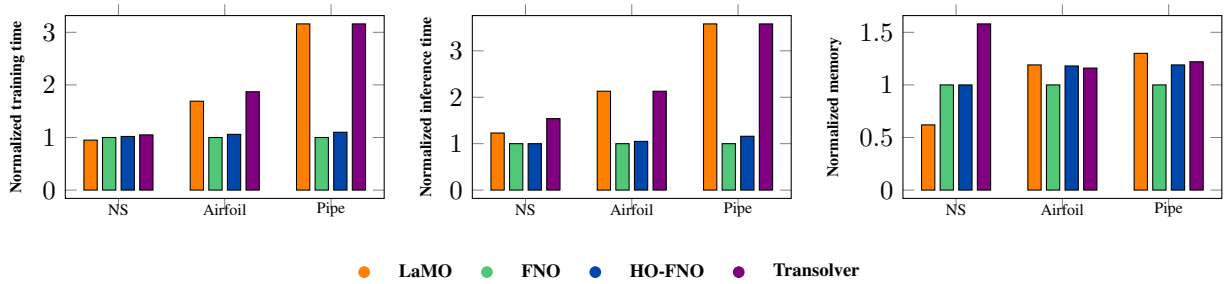

Figure 2. Efficiency comparison on NS, Airfoil, and Pipe after normalization with respect to FNO for each dataset and metric on a single Nvidia A100 GPU. A value of 1 corresponds to the computational cost of FNO; values below 1 indicate improved efficiency relative to FNO, whereas values above 1 indicate higher computational cost.

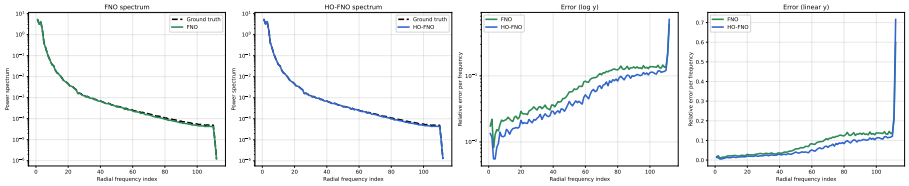

Figure 3. Airfoil

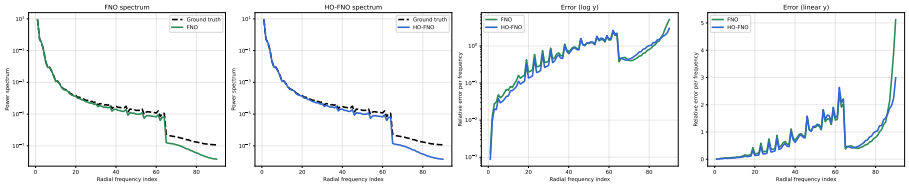

Figure 4. Pipe

Table 1. Main results across standard datasets, reported as mean relative ℓ_2 error (Eq. **??**; lower is better). Baseline numbers are taken from the corresponding table in (**?**) (same datasets, splits, and evaluation protocol), while **HO-FNO (Ours)** is evaluated by us under a parameter budget matched to LaMO on each task. In the color legend, orange denotes the best baseline, blue the second-best, and violet the third-best. Best result for each dataset is **in bold**.

	Operator	Res. equiv.	Regular Grid		Structured Mesh			Point Cloud
			Navier–Stokes	Darcy	Plasticity	Airfoil	Pipe	Elasticity
Classic	UNET (**?**)	✗	0.1982	0.0080	0.0051	0.0079	0.0065	0.0235
	Resnet (**?**)	✗	0.2753	0.0587	0.0233	0.0391	0.0120	0.0262
	SWIN (**?**)	✗	0.2248	0.0397	0.0170	0.0270	0.0109	0.0283
	DeepONet (**?**)	✗	0.2972	0.0588	0.0135	0.0385	0.0097	F
Transformer	Galerkin (**?**)	✗	0.1401	0.0084	0.0120	0.0118	0.0098	0.0240
	HT-Net (**?**)	✗	0.1847	0.0079	0.0333	0.0065	0.0059	/
	OFormer (**?**)	✗	0.1705	0.0124	0.0017	0.0183	0.0168	0.0183
	GNOT (**?**)	✗	0.1380	0.0105	0.0336	0.0076	0.0047	0.0086
	FactFormer (**?**)	✗	0.1214	0.0109	0.0312	0.0071	0.0060	/
	ONO (**?**)	✓	0.1195	0.0076	0.0048	0.0061	0.0052	0.0118
	Transolver (**?**)	✗	0.0957	0.0059	0.0013	0.0053	0.0046	0.0064
SSM	LaMO (**?**)	✗	0.0460	**0.0039**	0.0007	**0.0041**	**0.0038**	**0.0050**
Frequency	WMT (**?**)	✗	0.1541	0.0082	0.0076	0.0075	0.0077	0.0359
	U-FNO (**?**)	✓	0.2231	0.0183	0.0039	0.0269	0.0056	0.0239
	FNO (**??**)	✓	0.1556	0.0108	0.0074	0.0138	0.0067	0.0229
	U-NO (**?**)	✓	0.1713	0.0113	0.0034	0.0078	0.0100	0.0258
	F-FNO (**?**)	✓	0.2322	0.0077	0.0047	0.0078	0.0070	0.0263
	LSM (**?**)	✗	0.1535	0.0065	0.0025	0.0059	0.0050	0.0218
	FNO with improved backbone (Ours)	✓	0.0976	0.0068	0.0066	0.0065	0.0069	0.205
	HO-FNO (Ours)	✓	**0.0449**	0.0052	**0.0006**	0.0051	0.0052	0.0199
	Increment over FNO with improved backbone		**54.1%**	**21.2%**	**90.8%**	**26.1%**	**97.5%**	**90.3%**
	Increment over the best Frequency model		**70.7%**	**20.0%**	**76.0%**	**13.6%**	−4%	**8.7%**
	Increment over the best Transformer model		**53.1%**	**2%**	**53.8%**	**3.8%**	−13.0%	−211%
	Increment over the best Res. Equiv. model		**63.4%**	**46.2%**	**76.0%**	**82.4%**	**0.0%**	−68.6%

Table 2. Structure of the matrices A_s on the Darcy flow equation for HO-FNO of order 2 clarify : identity plus low-rank structure.

Structure	loss
Dense	5.2×10^{-3}
identity	7.4×10^{-3}
diagonal	7.0×10^{-3}
low-rank (128)	6.6×10^{-3}
low-rank (64)	7.5×10^{-3}
diagonal + low-rank (64)	7.2×10^{-3}
diagonal + low-rank (128)	6.4×10^{-3}
shared	6.7×10^{-3}

Dataset	Model	Training time (s)		Inference time (s)		Memory (GB)	
		Value	Δ (%)	Value	Δ (%)	Value	Δ (%)
NS	LaMO	117.10	-5.3	10.17	+22.8	28.45	-38.0
	FNO	123.63	0.0	8.28	0.0	45.85	0.0
	HO-FNO	125.81	+1.8	8.31	+0.4	45.94	+0.2
	Transolver	130.11	+5.2	12.72	+53.6	72.65	+58.5
Airfoil	LaMO	20.33	+69.2	1.77	+113.3	16.44	+18.7
	FNO	12.01	0.0	0.83	0.0	13.85	0.0
	HO-FNO	12.76	+6.2	0.87	+4.8	16.36	+18.1
	Transolver	22.45	+86.9	1.77	+113.3	16.06	+16.0
Pipe	LaMO	64.25	+215.9	5.62	+258.0	10.73	+30.4
	FNO	20.34	0.0	1.57	0.0	8.23	0.0
	HO-FNO	22.45	+10.4	1.82	+15.9	9.77	+18.7
	Transolver	64.25	+215.9	5.62	+258.0	10.07	+22.4

Table 3. check how the delta is defined and if the values are right ! Comparison of computational efficiency across datasets. For each metric, we report the absolute value and the relative variation (in %) with respect to FNO. Negative values indicate an improvement over FNO (lower is better), while positive values indicate a degradation.

	Navier-Stokes				Airfoil				Pipe			
	Full	Low	Medium	High	Full	Low	Medium	High	Full	Low	Medium	High
FNO	0.0976	0.145982	0.282080	0.458579	0.0065	0.025199	0.070703	0.146905	0.0069	0.191112	1.040052	1.253831
HO-FNO	0.0449	0.055541	0.206717	0.372876	0.0051	0.017370	0.045756	0.121412	0.0054	0.141260	1.031456	1.182214
Improvement over FNO	54.10%	61.95%	26.72%	18.69%	26.1%	31.07%	35.28%	17.35%	32.7%	26.1%	0.83%	5.71%

Table 4. Comparison of FNO and HO-FNO across datasets and regimes (Full, Low, Medium, High).

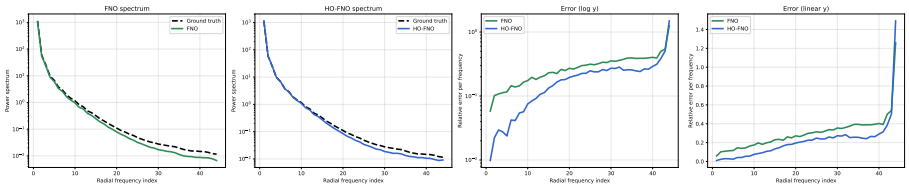

Figure 5. Navier-Stokes

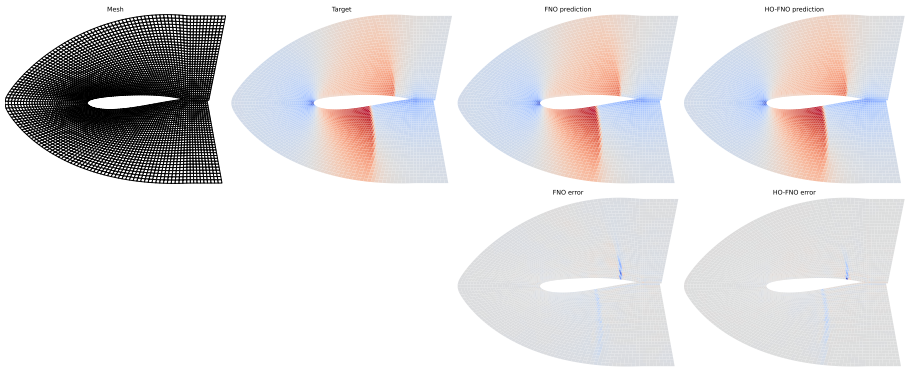

Figure 6. Airfoil

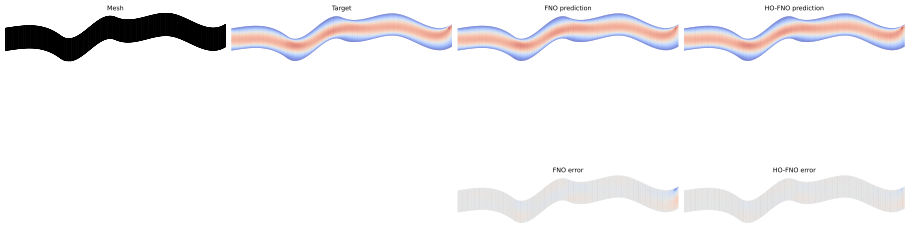

Figure 7. Pipe

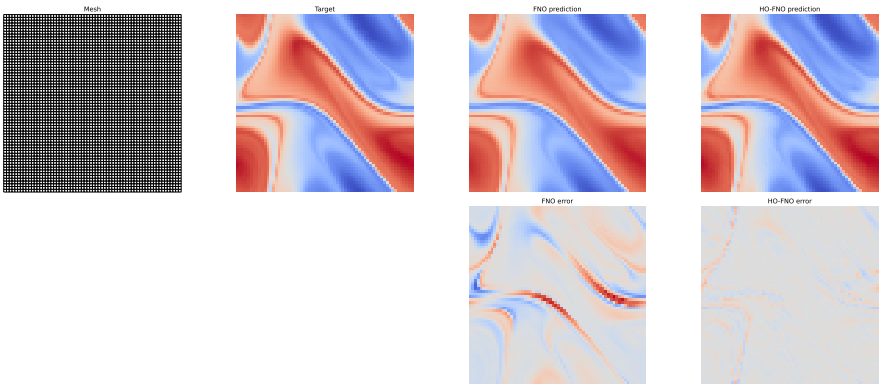

Figure 8. Navier-Stokes